Hypertensive retinopathy and Purtscher-like retinopathy in a child with complement 3 glomerulopathy

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We report the case of a 15-year-old boy with hypertensive retinopathy and Purtscher-like retinopathy eventually diagnosed with complement 3 glomerulopathy (C3G). The patient presented with bilateral severe painless visual loss and posterior pole cotton wool spots, optic disk and macular edema, and macular star-shaped hard exudate depositions, arterial narrowing, and venous tortuosity, indicative of hypertensive retinopathy (with an initial blood pressure of 210/130 mm Hg) and Purtscher-like retinopathy. He was subsequently diagnosed with C3G based on results of a kidney biopsy. There was a mild visual improvement on follow-up examination, and optic disk swelling and subretinal fluid and cotton wool spots resolved.

Complement 3 glomerulopathy (C3G) is a rare kidney disease caused by uncontrollable activation of the complement alternative pathway. It mainly affects older children, adolescents, and young adults. Patients typically present with hematuria, proteinuria, acute nephritis, nephrotic syndrome, hypertension, low levels of complement component C3, and renal failure.[1] Untreated secondary hypertension associated with C3G and changes in complement system activity can cause ocular manifestations. Occlusive chorioretinopathy has been reported in a number of systemic disorders as a result of overactivation of the complement system.[2] We report the case of a child with severe vision loss and ophthalmic features of hypertensive chorioretinopathy and Purtscher-like retinopathy who was eventually diagnosed with C3G.

Case Report

A 15-year-old boy presented at the ophthalmology clinic of Khatam Al-Anbia Eye Hospital with bilateral painless vision loss of 2 months' duration. His parents reported a history of nephrotic syndrome treated with oral prednisolone 2 years before presentation, but the patient had stopped taking his medication for over a year and had had no recent medical evaluation. There was no history of a recent infectious disease. Review of systems did not reveal any associated symptoms (eg, headache, hematuria).

On examination, best-corrected visual acuity was 20/100 in the right eye and counting fingers at 1 m in the left eye. A relative afferent pupillary defect of 2+ was detected in the left eye. Intraocular pressure measured 14 mm Hg in the right eye and 13 mm Hg in the left eye. Anterior segment examination of both eyes was unremarkable. Fundus examination of both eyes demonstrated optic disk edema, subretinal fluid, arterial narrowing, and extensive cotton wool spots all over the posterior pole and involving the peripapillary area, as well as vascular tortuosity, and a few dot and blot hemorrhages, hard exudate deposition in a star-shaped configuration, and multiple peripheral hypopigmented choroidal spots (Figure 1). Spectral domain optical coherence tomography (SD-OCT) showed irregularity and wrinkling of the retinal surface, retinal thickening, subretinal and intraretinal fluid, and hyperreflective deposits with posterior shadowing compatible with hard exudates in the inner retinal layers (eSupplement 1, available at jaapos.org). Optic nerve head OCT (ONH-OCT) was indicative of increased peripapillary retinal thickness. Fundus autofluorescence showed areas of hyperautofluorescence in the macular region, compatible with subretinal fluid. Also, hypoautofluorescence was found to correspond to areas of retinal whitening (eSupplement 2, available at jaapos.org). On systemic evaluation, blood pressure was 210/130 mm Hg. Because the patient was diagnosed with secondary hypertension based on his age, he was referred to the pediatric ward for complete evaluation. The results of laboratory tests are provided in Table 1; other examinations, such as liver function tests, serum electrolytes, and coagulation tests were within normal ranges. The patient was admitted to the intensive care unit to manage hypertension and renal failure. Mild left ventricular hypertrophy, mild mitral regurgitation, and mild aortic insufficiency were found on echocardiography. The only pathologic findings on abdominopelvic sonography were loss of corticomedullary differentiation and increased echogenicity of both kidneys, which suggested parenchymal kidney disease. A renal biopsy demonstrated diffuse proliferative glomerulonephritis and predominant C3 staining on immunofluorescence consistent with C3G. The patient's blood pressure was controlled during his admission, and methylprednisolone pulse therapy and mycophenolate mofetil were prescribed. A mild subjective improvement in visual acuity following improved blood pressure control was noted over the next few days. He was discharged on antihypertensive medications (captopril, amlodipine), mycophenolate mofetil, and oral prednisolone (12.5 mg daily).

At his 2-week follow-up visit, best-corrected visual acuity in the right eye had improved to 20/50; the left eye

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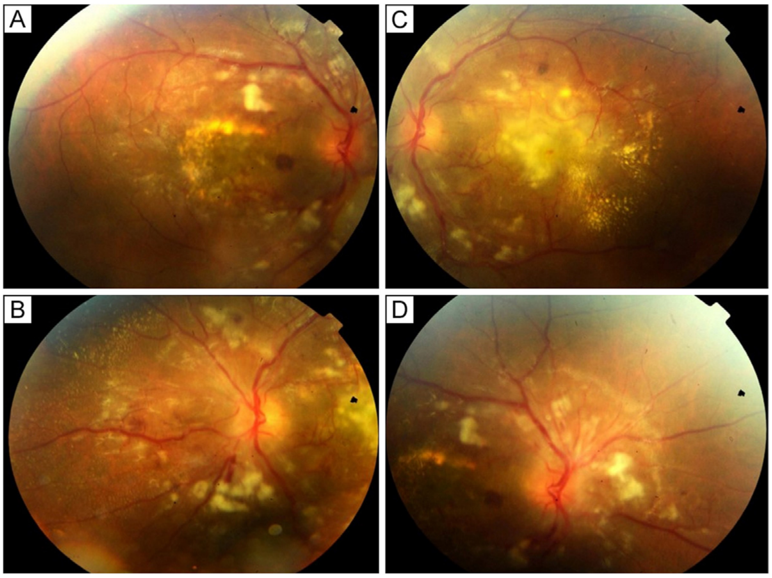

FIG 1. Fundus photography (right eye, A-B; left eye, C-D) showing optic disk edema, exudative retinal detachment, arterial narrowing, extensive cotton wool spots all over the posterior pole involving the peripapillary area, as well as venous tortuosity, a few dot and blot hemorrhages, and star-shaped hard exudate deposition.

remained unchanged. According to fundus examination and OCT images, optic disk swelling, subretinal fluid and peripapillary cotton wool spots were diminished (Figure 2).

Rituximab was prescribed by his pediatrician based on his impaired renal function during follow-up. Due to extensive glomerular damage, the renal impairment did not respond to medications, and hemodialysis was administered weekly.

At 18 weeks' follow-up, visual acuity had improved to 20/25 in the right eye and to counting fingers at 3 m in the left eye. Follow-up imaging revealed improvement in fundus abnormalities, with resolution of macular edema and optic disk swelling, but residual outer retinal hyperreflective foci in both eyes, retinal pigment epithelium (RPE) irregularity, especially in the left eye, and destruction of the inner/outer segment in the left eye (eSupplement 3, available at jaapos.org).

Table 1. Laboratory test results

Test	Patient value	Normal value (ranges)
Hemoglobin, g/dl	11	13-18
Hematocrit, %	31	40-54
Platelets, ×103/μl	208	150-450
Leukocytes, ×103/μl	9.4	4-10
Albumin, mg/dl	3.6	3.5-5.2
Creatinine, mg/dl	3.2	0.22-0.59
Cholesterol, mg/dl	226	<200
Triglycerides, mg/dl	232	40-200
Urinalysis, mg/dl		
Random urine protein	197	Negative
Random urine creatinine	16.9	
Red blood cell	2-3/high power field	

Discussion

C3G represents a rare group of kidney disorders characterized by the dysregulation of the complement alternative pathway in the fluid phase. In the glomerular microenvironment, this is indicated by the presence of complement C3 deposition on kidney biopsy specimens.[3] Symptoms of C3G can range from asymptomatic hematuria and proteinuria to an acute presentation with glomerulonephritis. In addition to hematuria and hypertension, patients may also have a history of acute kidney injury or chronic kidney disease. A low serum C3 level is a common indicator of C3G and is often the first sign of the condition.[4] In accordance with the pathophysiology of C3G, the disorder can cause multiple ocular manifestations. Uncontrolled secondary hypertension caused by C3G may lead to hypertensive chorioretinopathy. Overactivation of the complement system may also lead to the deposition of complement system products in the RPE, manifested as retinal drusen.[5] Occlusive retinopathy has also been reported in C3G.[6]

Our patient's severe hypertension, in addition to optic disk edema and signs of choroidal ischemia, suggested hypertensive retinopathy grade IV. However, optic disk edema in hypertensive retinopathy is typically less severe and associated with more flame-shaped hemorrhages.

Miguel and colleagues[2] have suggested that at least three of the following criteria are necessary to diagnose Purtscher retinopathy: Purtscher flecken, a low-to-moderate number of hemorrhages, cotton wool spots (typically restricted to posterior pole as here), and probable or plausible explanatory etiology (renal disease and complement dysfunction are present here). Because our patient has both Purtscher-like and hypertensive retinopathy features, it is likely that both diseases occurred

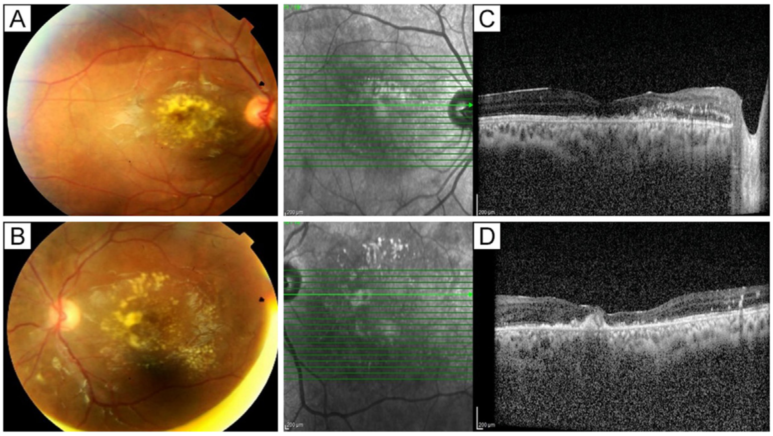

FIG 2. Color fundus photography of both eyes (A-B) at 2 weeks' follow-up showing nearly complete resolution of optic disk edema, hemorrhages, and Purtscher flecken. Follow- up macular OCT of both eyes (C-D) showing resolution of cystoid macular edema and subretinal fluid, with normalization of the foveal contour, with persistent outer retinal abnormalities and outer retinal hyperreflective foci in both eyes.

simultaneously. Although Purtscher retinopathy can cause optic disk and macular edema, which may be associated with poor visual outcomes,[2] the presence of choroidal infarctions and the initial severe optic disk edema may not be fully explained by Purtscher retinopathy.

It has been well documented that renal disease and Purtscher-like retinopathy are associated.[7] There is only one report of Purtscher-like retinopathy and hypertensive retinopathy in association with C3G.[6] Our case is a rare presentation of simultaneous hypertensive retinopathy and Purtscher-like retinopathy in relation to C3G in a child. One noteworthy aspect of this case is the simultaneous presence of several ocular manifestations of C3G, including subretinal drusenoid deposits, evidence of Purtscher-like retinopathy, and hypertensive retinopathy.

The differential diagnosis for a child with optic disk edema and star-shaped macular exudate includes neuroretinitis, which may be caused by infection, and hypertensive retinopathy.[8] Our patient's history and initial investigations did not reveal any signs of infection. Secondary hypertension is more common than primary hypertension in children, and renovascular disease is one of the most common causes of secondary hypertension. Other causes of secondary hypertension must also be considered such as antiphospholipid syndrome, aldosteronism, or pheochromocytoma.[9] Early diagnosis and coordination of pediatric ophthalmic care can reduce the risk of long-term complications.

There is no complete understanding of the pathophysiology of Purtscher-like retinopathy, but physical examination and fluorescein angiography findings suggest embolic occlusion of precapillary arterioles. Emboli may consist of air, fat, platelets, fibrin, or leukocytes, and may occur in the context of an overactive complement system.[10] Complement activation can contribute to Purstcher-like retinopathy directly by activating leukocyte or platelet aggregations or indirectly by damaging endothelial cells and triggering the clotting cascade.[10] C3 glomerulopathy currently lacks a proven treatment.[3] In addition to renoprotective measures, immune-suppressive medications are occasionally prescribed, and new complement inhibitors are being investigated.[11] Eculizumab, a C5 inhibitor, has recently shown some efficacy in the treatment of C3G, although it has not been studied in a clinical trial. Several other new anticomplement therapies are under development.[11]

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